

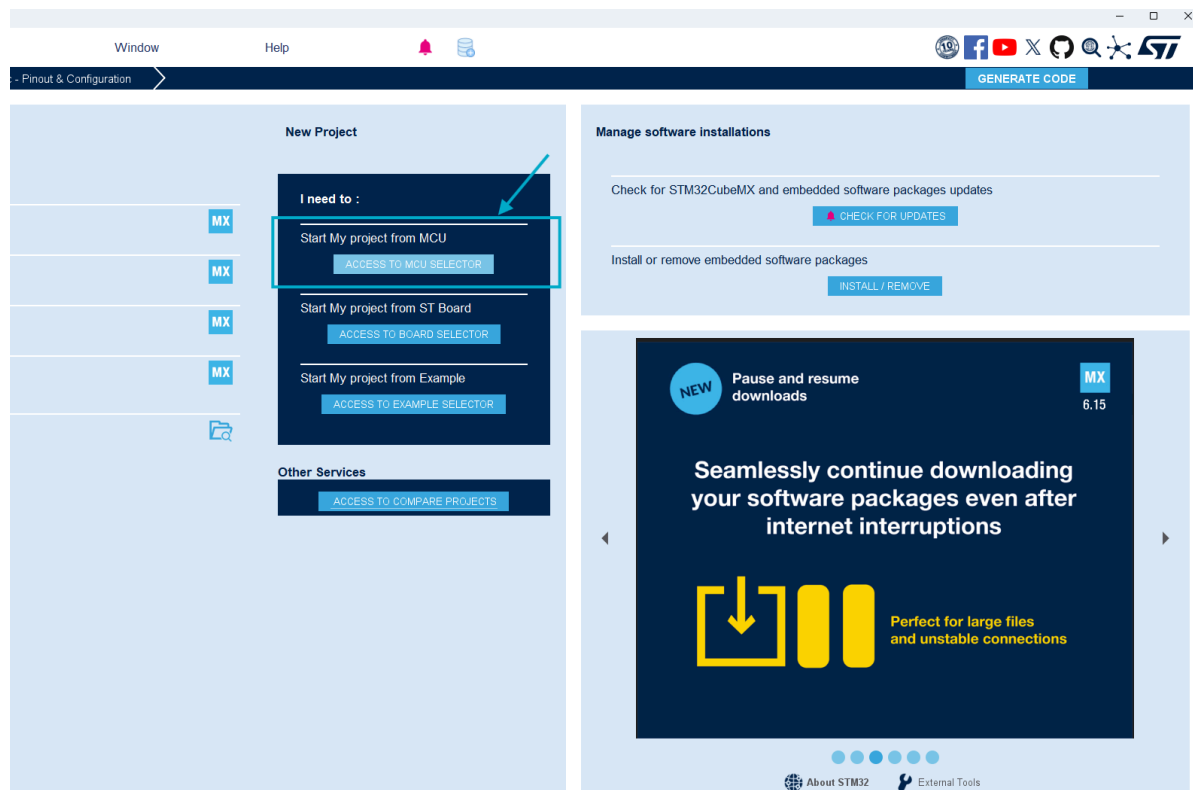
本文档是整个cube + EIDE工作流配置的教程，每次新建工程都会重复这样的工作，建议熟练掌握

参考文档：[note/EC/EIDE工作流.md at main · lynliam/note · GitHub](#)

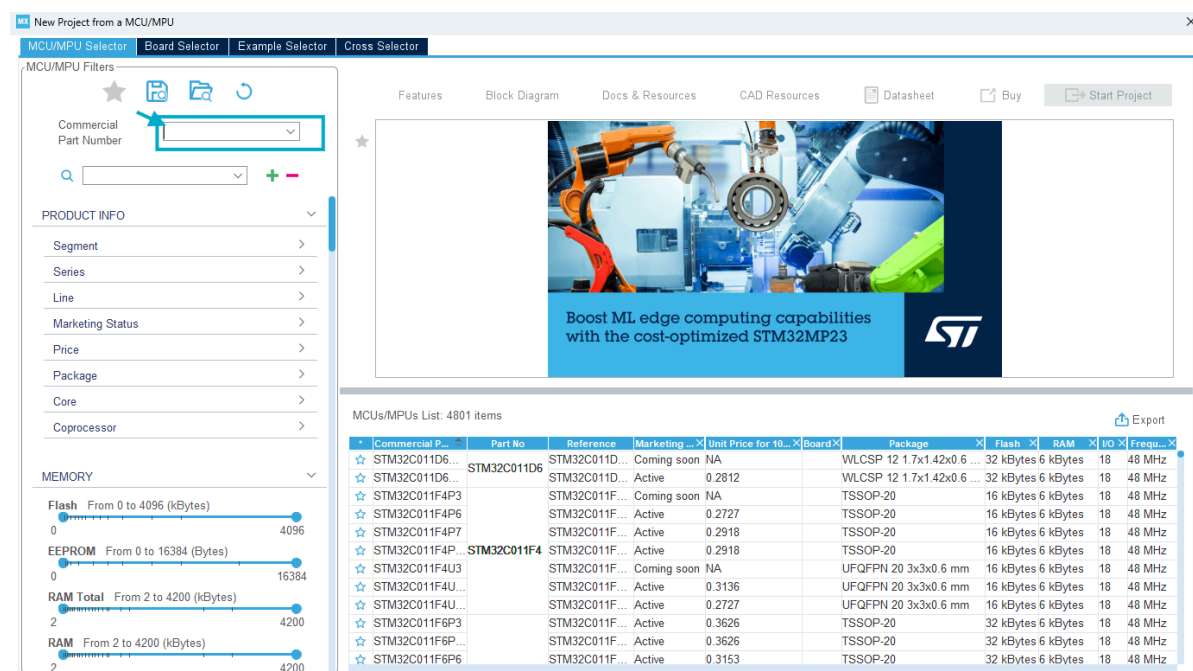
CubeMX配置方法

1.芯片选型

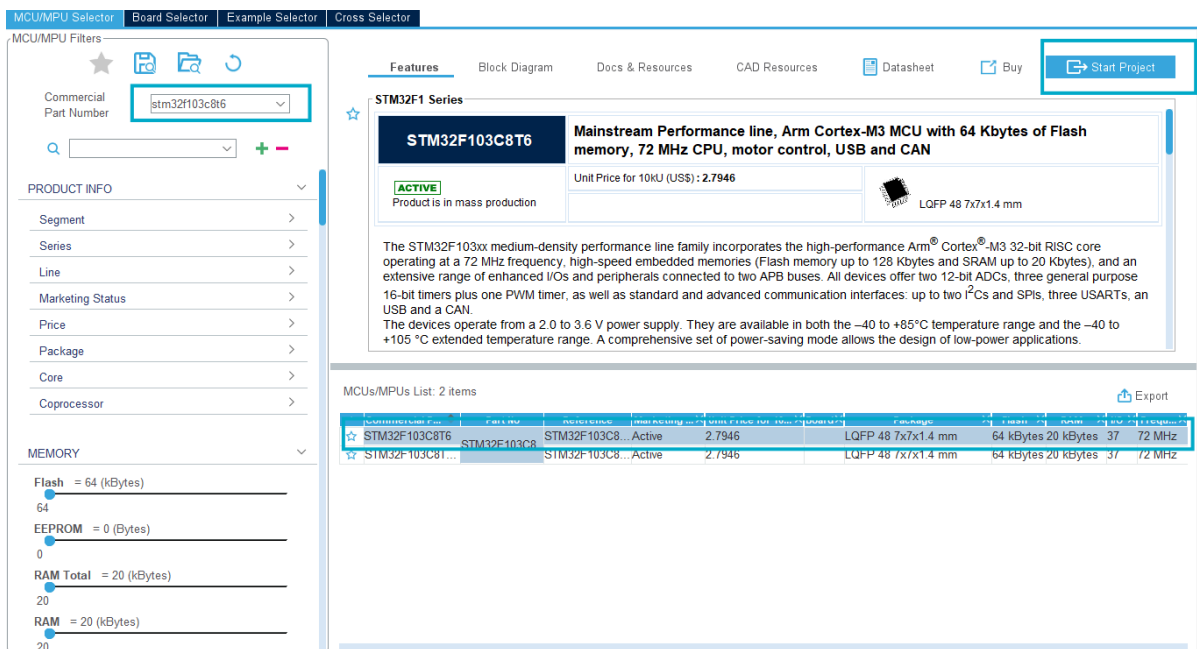
打开Cubemx，左侧一栏为历史工程，中间这一栏是新建工程。我们一直都是用下图框选的这个选项，点击ACCESS TO MCU SELECTOR,选择芯片型号



会弹出来这个界面，在搜索框里搜索stm32f103c8t6, 这是我们培训期间使用的板子型号

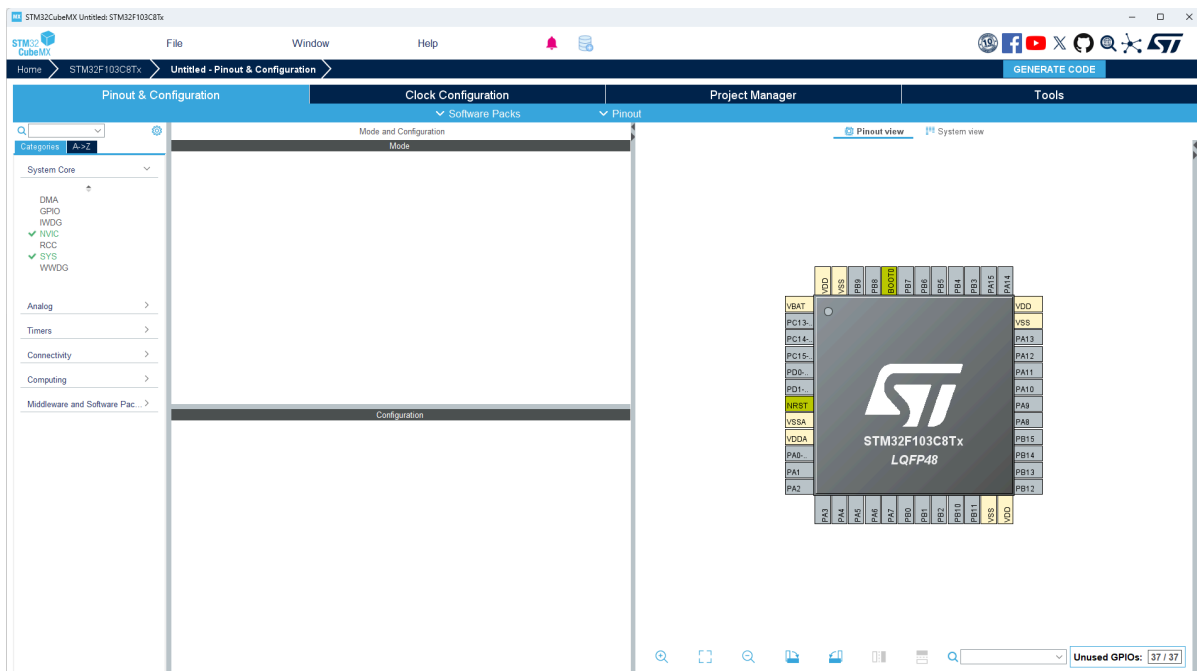


右图的芯片型号栏里剩下两个芯片，点击第一个，再点击右上角的Start Project (或者双击两次板子型号也可以直接开启项目)

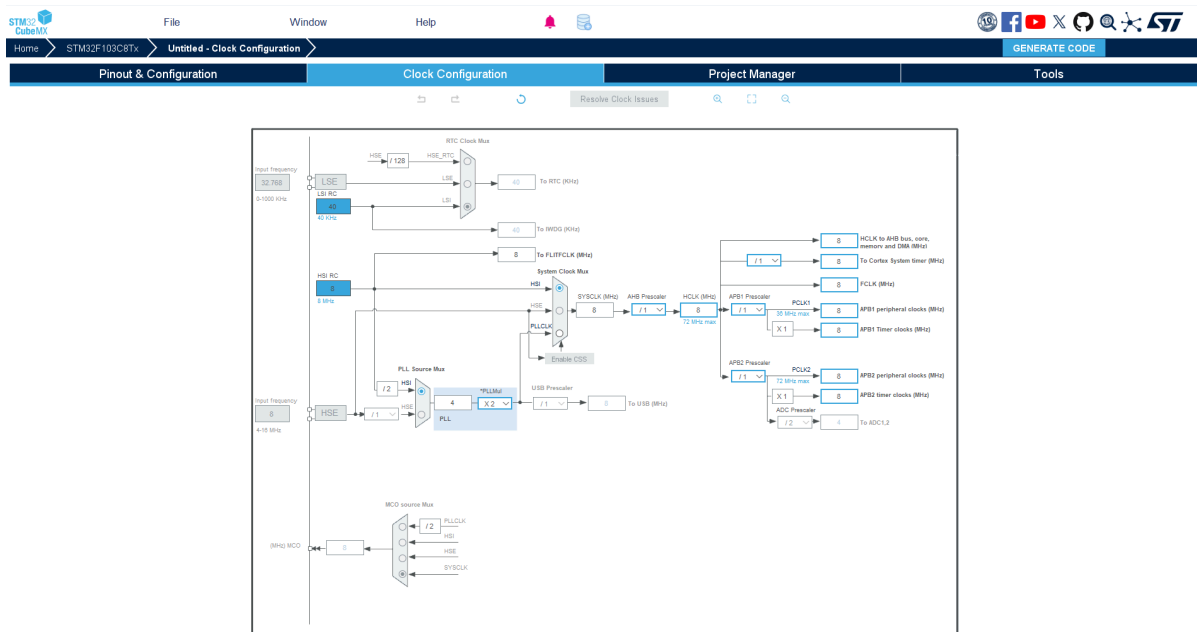


开启工程会见到如下界面（简单介绍一下，看不懂没关系，后面用着用着就熟了）

Pinout&Configuration:左边这一栏是系统核心和外设的列表，中间这一栏是详细配置信息，右边是芯片的的引脚图，可视化地呈现了对每一个引脚的配置



Clock Configuration:时钟树的配置



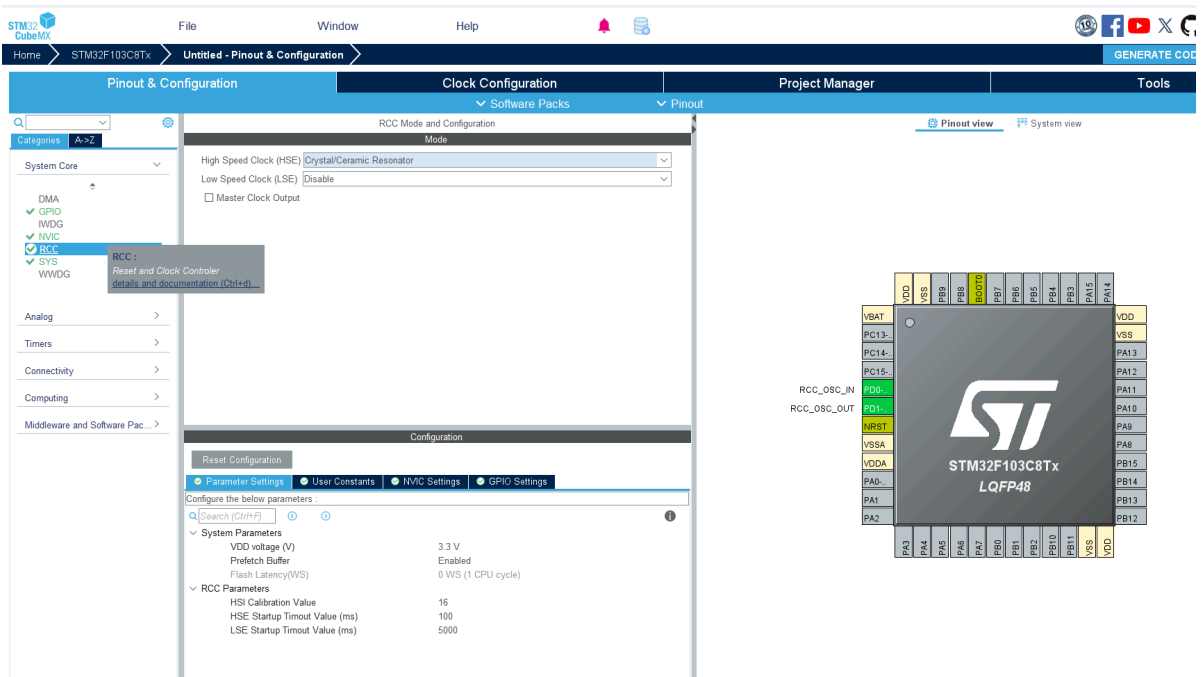
Project Manager:配置项目信息，包括项目命名等

Pinout & Configuration	Clock Configuration	Project Manager	Tools
Project	Project Settings Project Name Project Location Application Structure Toolchain Folder Location		
Code Generator	Toolchain / IDE Min Version Generate Under Root		
Advanced Settings	Linker Settings Minimum Heap Size Minimum Stack Size		
	Thread-safe Settings Cortex-M3NS Enable multi-threaded support Thread-safe Locking Strategy		
	Mcu and Firmware Package Mcu Reference Firmware Package Name and Version Use latest available version Use Default Firmware Location Firmware Relative Path		

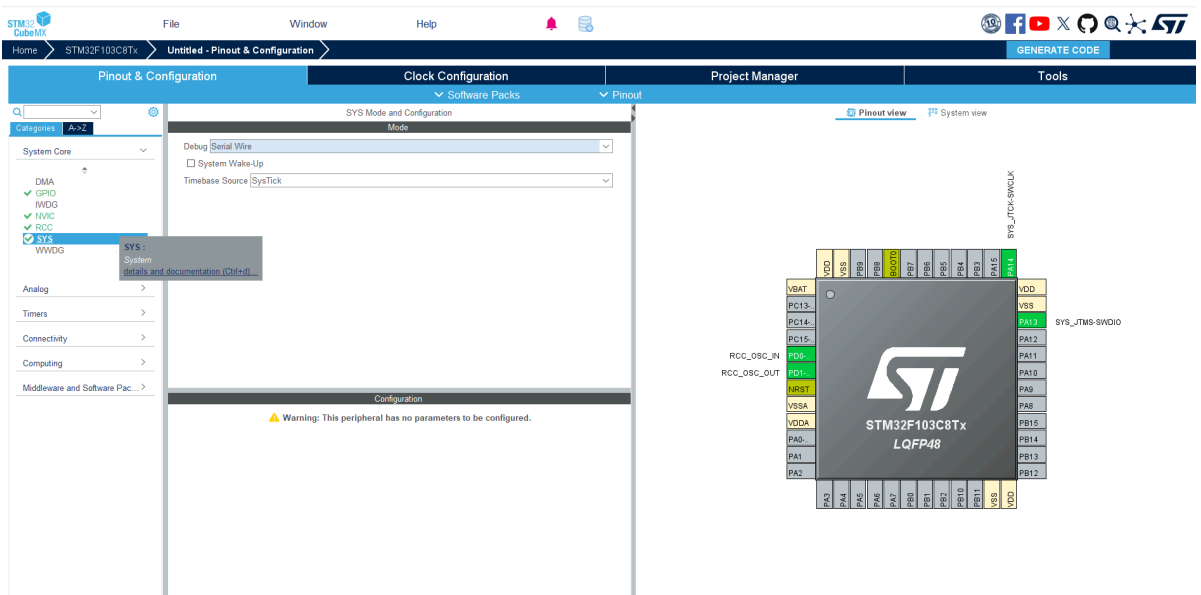
2.基础配置

接下来是重点：CubeMX的一些基础配置，就是后面每一次配置新工程都必走的流程

左栏点击RCC，High Speed Clock选择Crystal/Ceramic Resonator，意思是为高速时钟源选择晶振

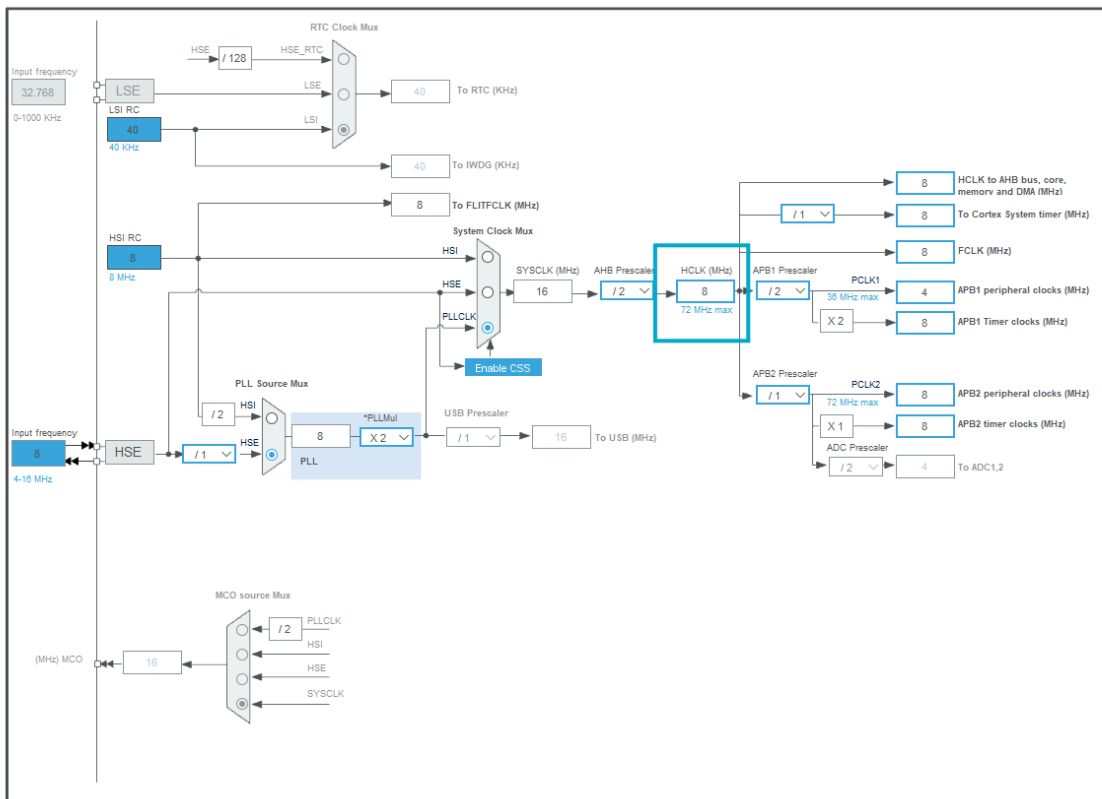


左栏点击RCC, Debug选择Serial Wire, 选择调试模式为SWD模式

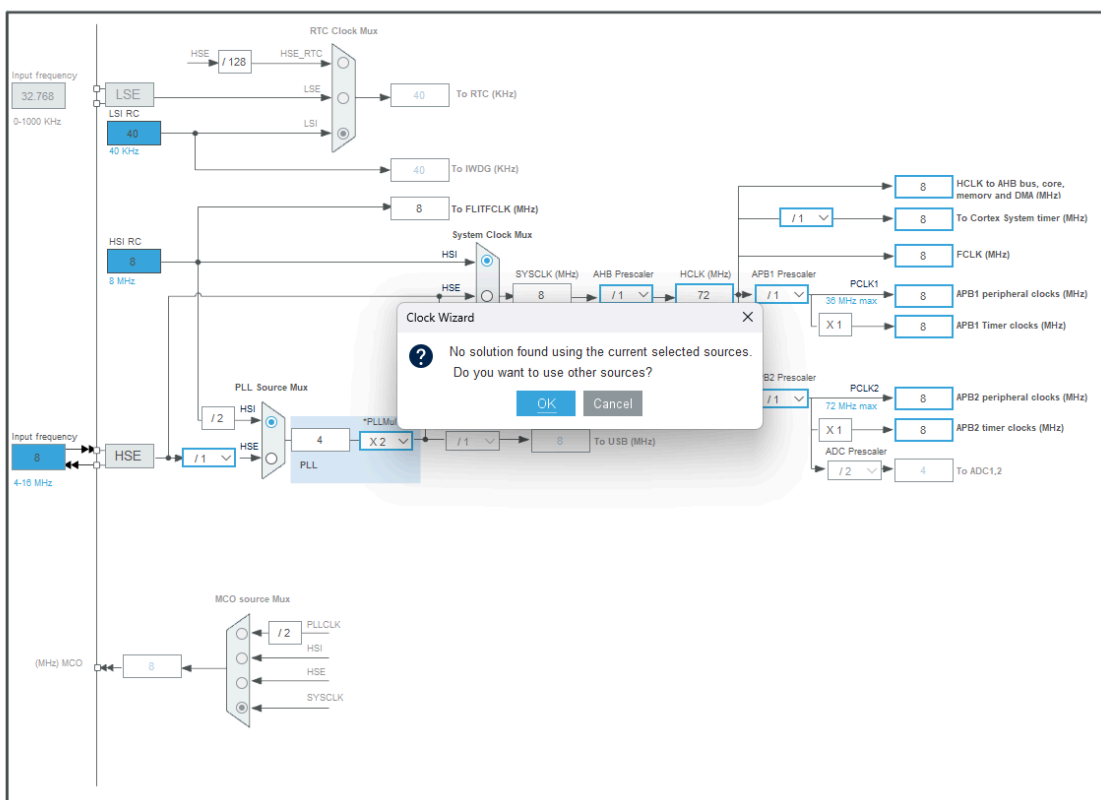


然后来到时钟树的配置：

把下框的数字8改为72，回车



弹出这个界面就选OK。这里配置的是芯片运行主频，c8t6的最高主频是72MHz, 我们拉到最大就好



然后来到Project Manager,配置自己习惯的路径, 不要含中文不要含中文

STM32CubeMX

File Window Help

Home > STM32F103C8Tx > Untitled - Project Manager

GENERATE CODE

Pinout & Configuration Clock Configuration Project Manager Tools

Project

Project Settings

Project Name ITS_peixun 输入项目名称, 不要含中文

Project Location F:\VTS_Log\Workspace\stm32_workspace 选择项目存放位置, 不要乱放以免自己找不到, 不要含中文

Application Structure Advanced ☐ Do not generate the main()

Code Generator

Toolchain Folder Location F:\VTS_Log\Workspace\stm32_workspace\ITS_peixun\

Toolchain / IDE Makefile 选择Makefile模式 ☐ Generate Under Root

Advanced Settings

Linker Settings

Minimum Heap Size 0x200

Minimum Stack Size 0x400

Thread-safe Settings

Cortex-M3NS

☐ Enable multi-threaded support

Thread-safe Locking Strategy Default - Mapping suitable strategy depending on RTOS selection

Mcu and Firmware Package

Mcu Reference STM32F103C8Tx

Firmware Package Name and Version STM32Cube_FW_F1_V1.8.6 Use latest available version

☒ Use Default Firmware Location

Firmware Relative Path C:\Users\zyx\STM32CubeRepository\STM32Cube_FW_F1_V1.8.6

STM32CubeMX

File Window Help

Home > STM32F103C8Tx > ITS_peixun.ioc - Project Manager

GENERATE CODE

Pinout & Configuration Clock Configuration Project Manager Tools

Project

STM32Cube MCU packages and embedded software packs

☐ Copy all used libraries into the project folder

☒ Copy only the necessary library files

☐ Add necessary library files as reference in the toolchain project configuration file

Code Generator

Generated Files

☒ Generate peripheral initialization as a pair of c/h files per peripheral

☐ Backup previously generated files when re-generating

☒ Keep User Code when re-generating

☒ Delete previously generated files when not re-generated

Advanced Settings

HAL Settings

☐ Set all free pins as analog (to optimize the power consumption)

☐ Enable Full Assert

OK,点击右上角的GENERATE CODE

STM32CubeMX

File Window Help

Home > STM32F103C8Tx > Untitled - Project Manager

GENERATE CODE

Pinout & Configuration Clock Configuration Project Manager Tools

Project

STM32Cube MCU packages and embedded software packs

☒ Copy all used libraries into the project folder

☐ Copy only the necessary library files

☐ Add necessary library files as reference in the toolchain project configuration file

Code Generator

Generated Files

☒ Generate peripheral initialization as a pair of c/h files per peripheral

☐ Backup previously generated files when re-generating

☒ Keep User Code when re-generating

☒ Delete previously generated files when not re-generated

Advanced Settings

HAL Settings

☐ Set all free pins as analog (to optimize the power consumption)

☐ Enable Full Assert

User Actions

Before Code Generation

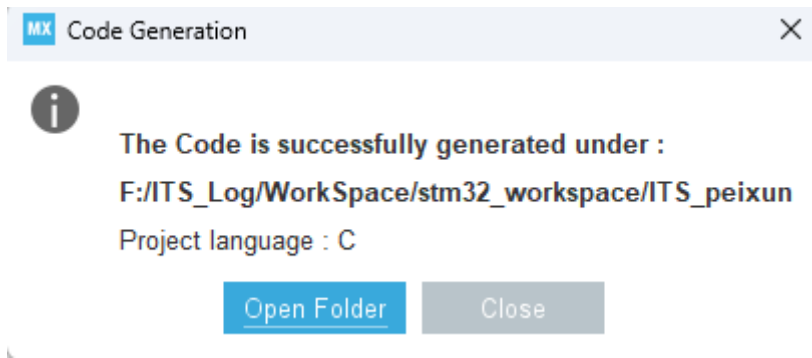
After Code Generation

Template Settings

Return to template to generate customized code

第一次使用该型号芯片时会出现让你下载相应压缩包文件的弹窗，按照它的指示下载、注册ST账号即可，如果遇到一些网络问题就来战队下载

最后代码生成完毕之后会弹出这个，close即可。如果感兴趣也可以openfolder看一下

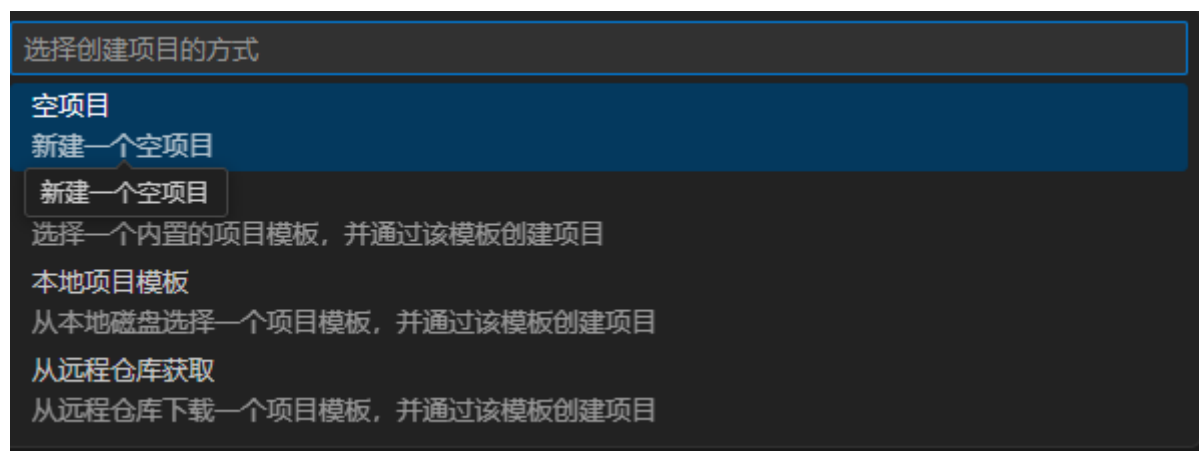


EIDE配置

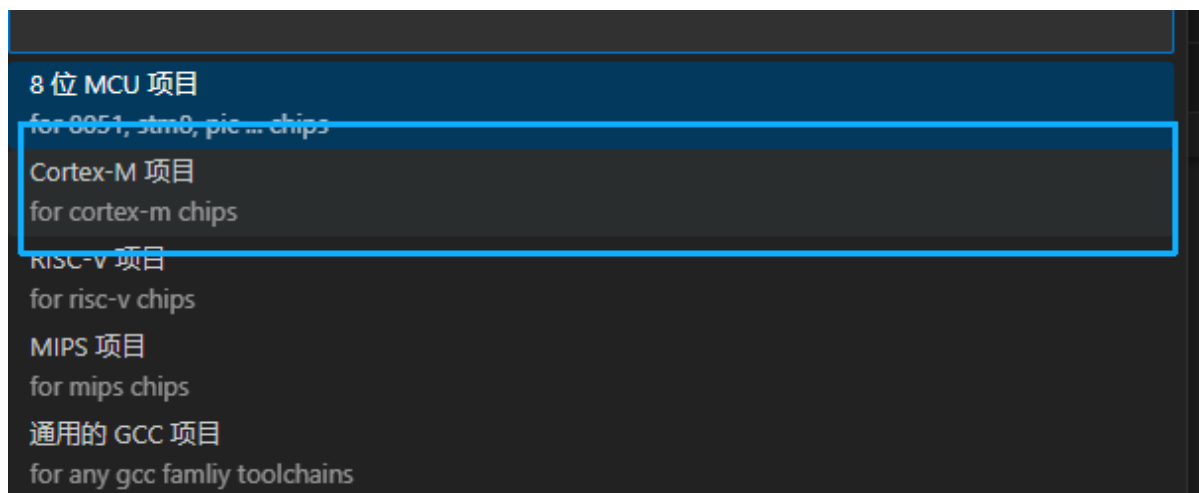
打开Vscode，点击左侧工具栏EIDE图标，新建项目



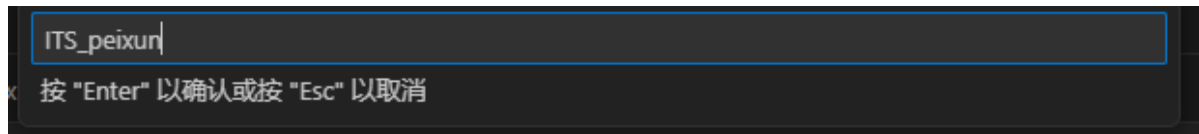
选择空项目



然后选择Cortex-M项目

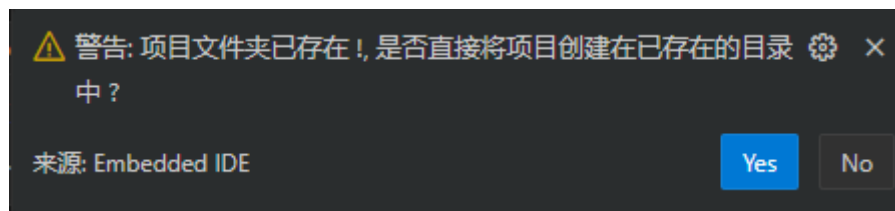


它会让你输入项目名称，记住这里的命名一定一定要和刚才在cube填入的工程名称一模一样，如 ITS_peixun， 敲击Enter

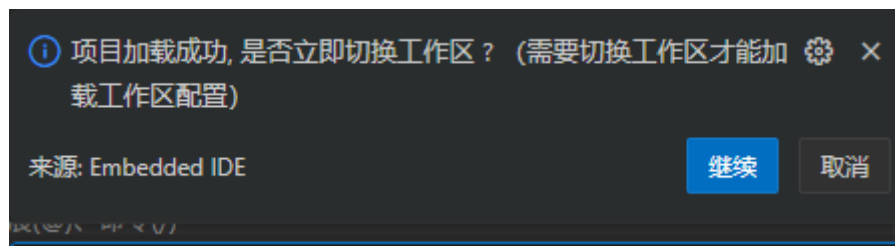


然后会让你选择文件存放位置，要存放在项目的上一级目录，比如我的整个路径是 F:\ITS_Log\WorkSpace\stm32_workspace\ITS_peixun，那我这里就应该选择放在 stm32_workspace底下，因为这里其实是让EIDE创建的项目复写原来的Cube生成的代码文件，如果路径选错了会得到很抽象的工程文件，建议全删了重新来过

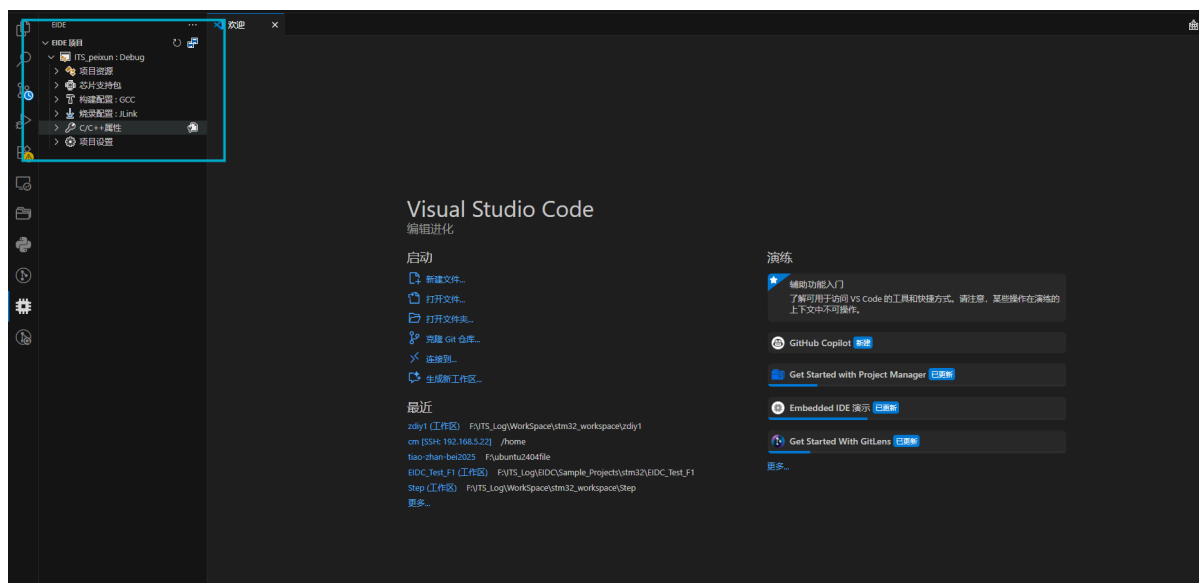
如果路径选对了，右下角会出现这个弹窗，说明复写是成功的，点Yes！！（没有出现弹窗的乖乖去重配好了）



还是在右下角，会继续出现这个弹窗，选择继续即可

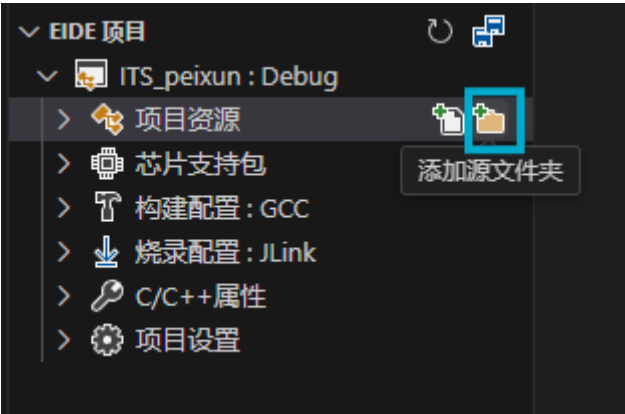


然后就会转换到这个工作区（没有的话点一下左边的eide图标

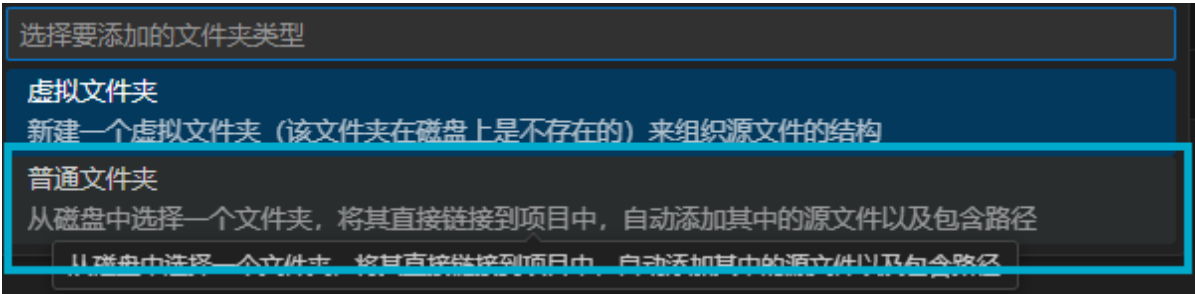


接下来是eide项目的配置

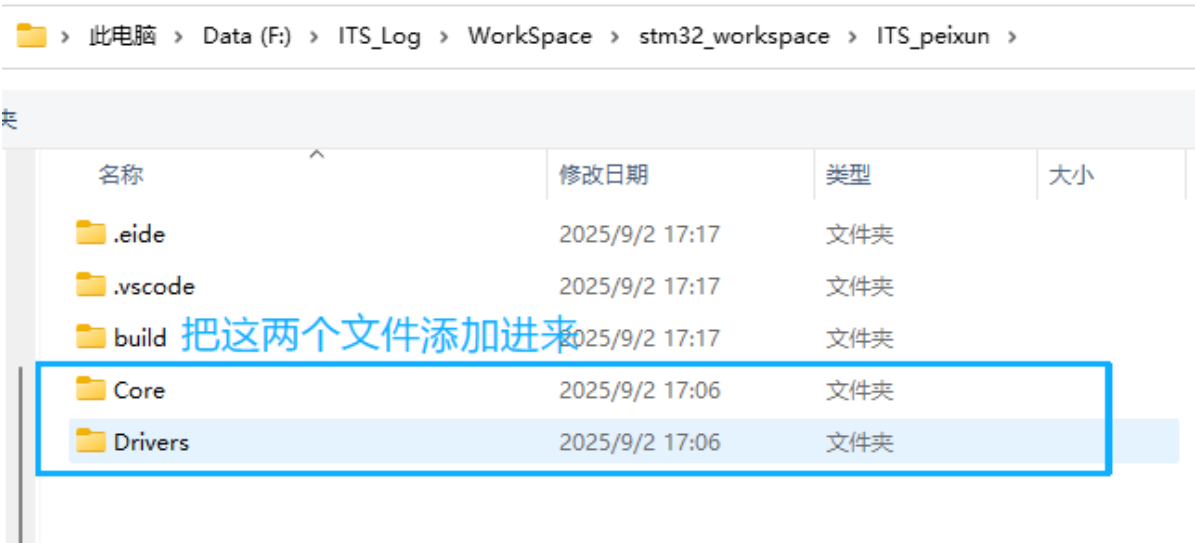
添加源文件夹：



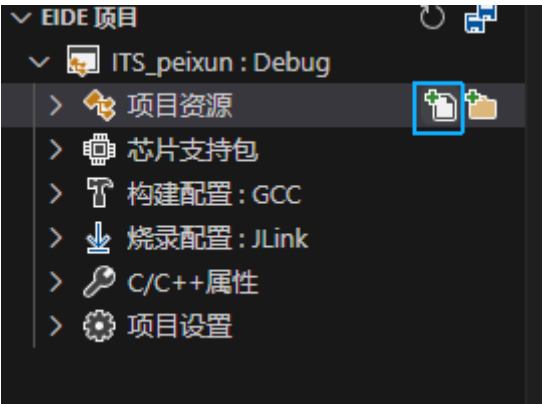
选择普通文件夹：



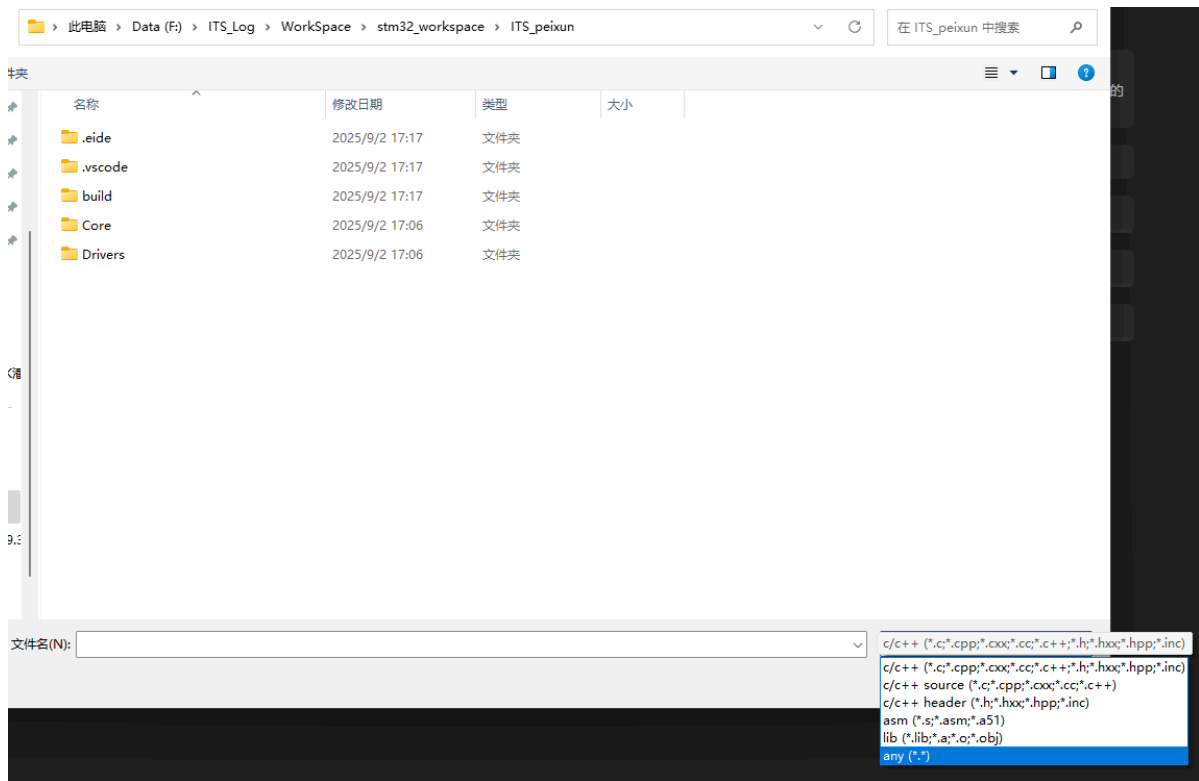
选择下框两个文件夹，点击添加原文件夹到项目



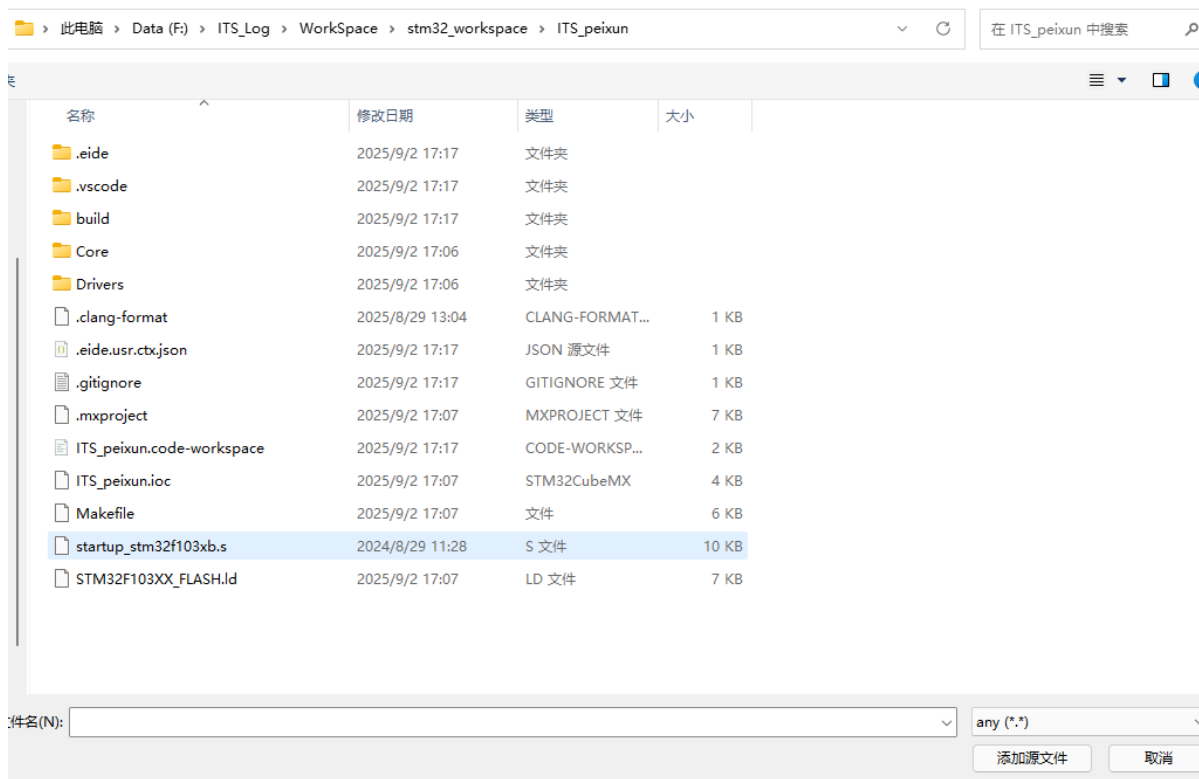
接续添加源文件，选择这个图标



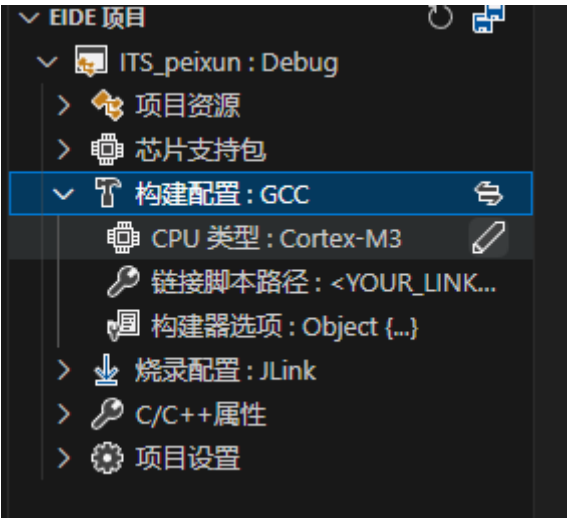
然后把下面的显示文件种类选为any或asm



可以看到多出了很多文件，把下图所选的哪个.s文件添加进来



然后是构建配置：stm32f103c8t6的CPU类型是Cortex-M3没问题

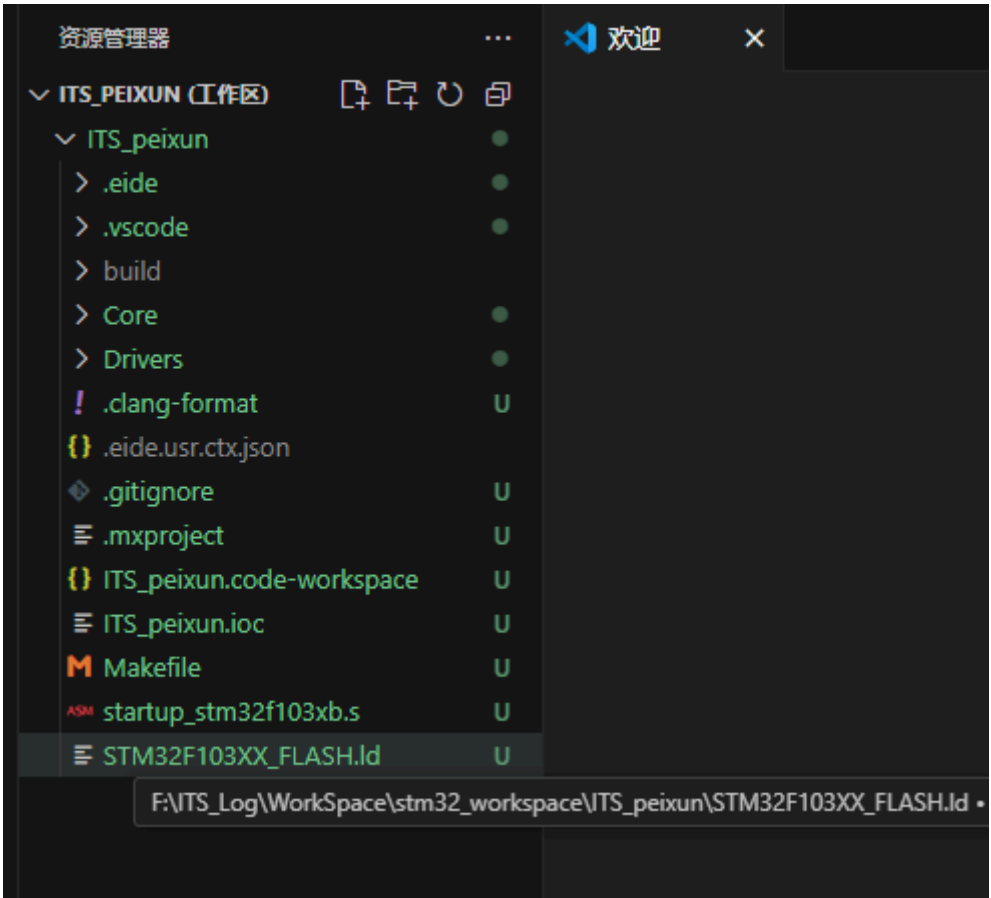


链接脚本路径点击这个图标

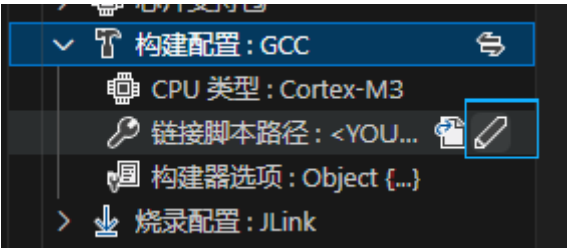


打开资源管理器

看到最后那个STM32F103的.ld文件，右键它选择复制相对路径



然后回到刚才的地方，点这个修改

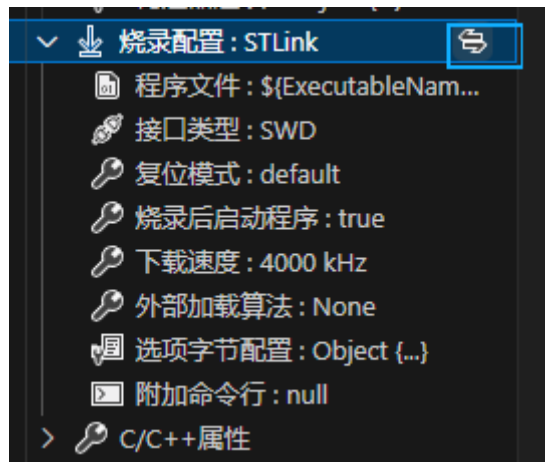


直接ctrl+v复制进来，回车即可

STM32F103XX_FLASH.ld

按 "Enter" 以确认或按 "Esc" 以取消

烧录配置：点击下框按钮切换烧录器为STLink



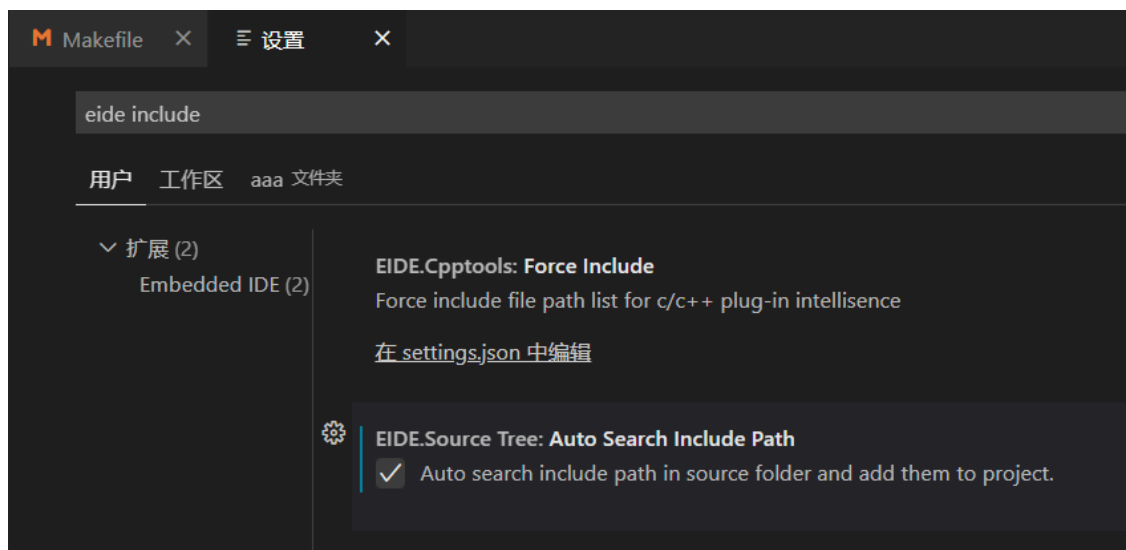
C/C++属性:

包含目录

(==强烈建议==) 如果设置中勾选了 `EIDE.Source Tree: Auto Search Include Path` (如下图), 那么 EIDE 会自动搜索项目资源中添加的文件夹, 这里就不用手动填入了

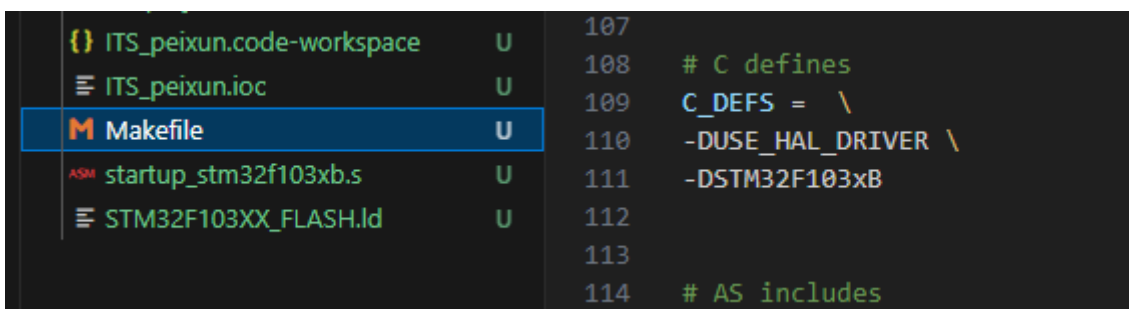
打开插件设置



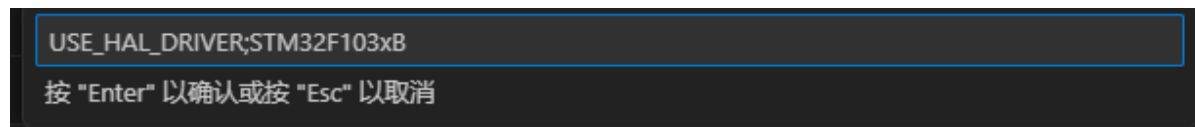
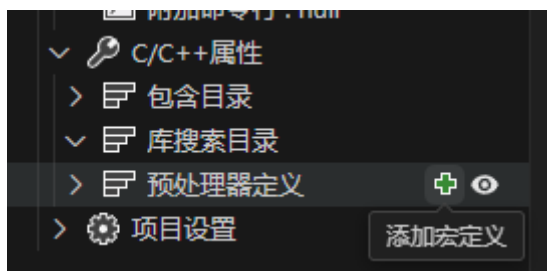


添加预处理器定义

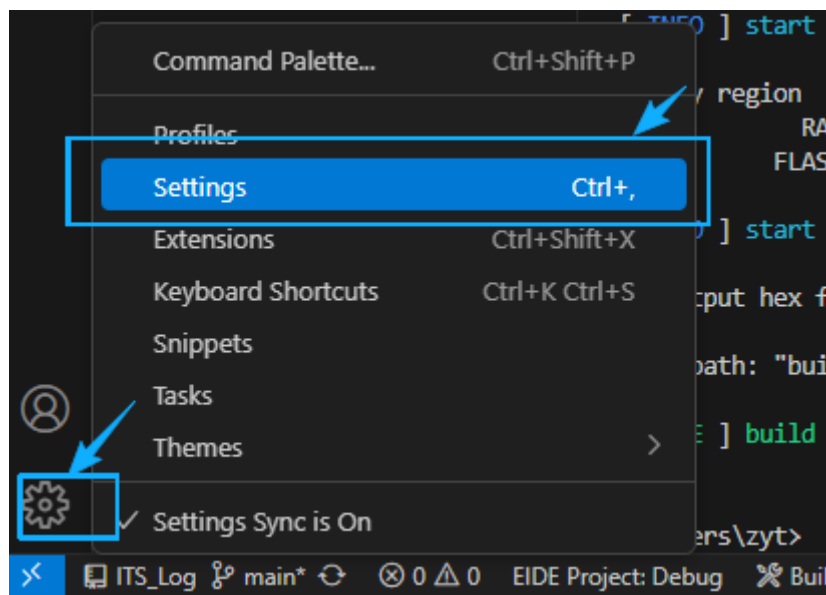
打开Makefile，在一百来行这里有个CDEFS，把这两行复制下来，-D删掉，中间用一个分号连接变成这样：USE_HAL_DRIVER;STM32F103xB，复制下来



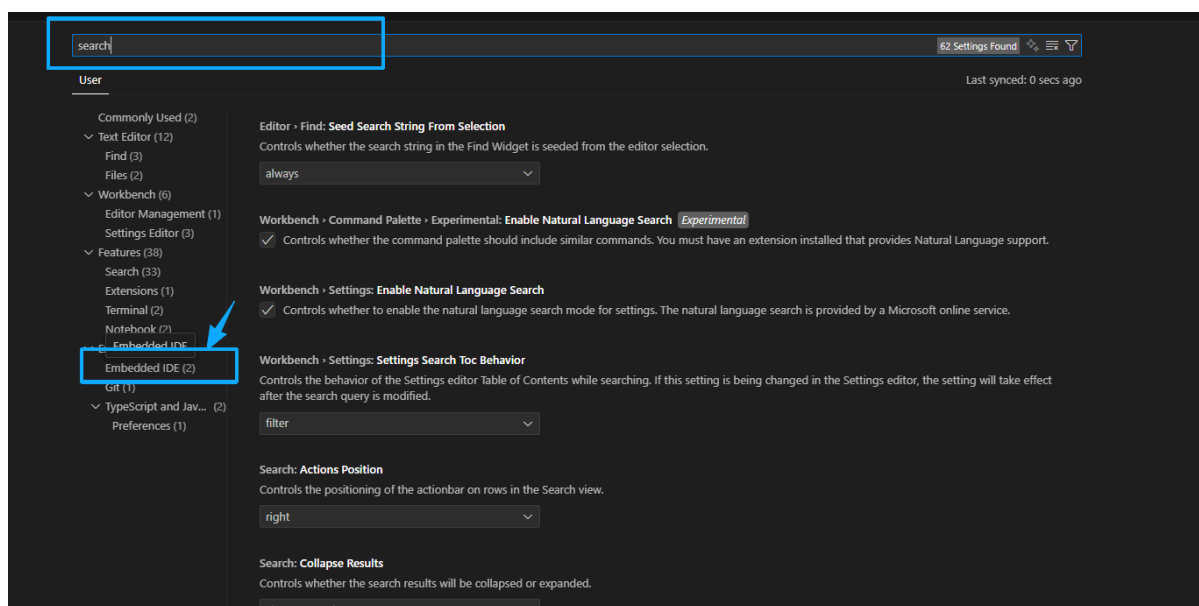
点击这个绿色的加号，复制，回车



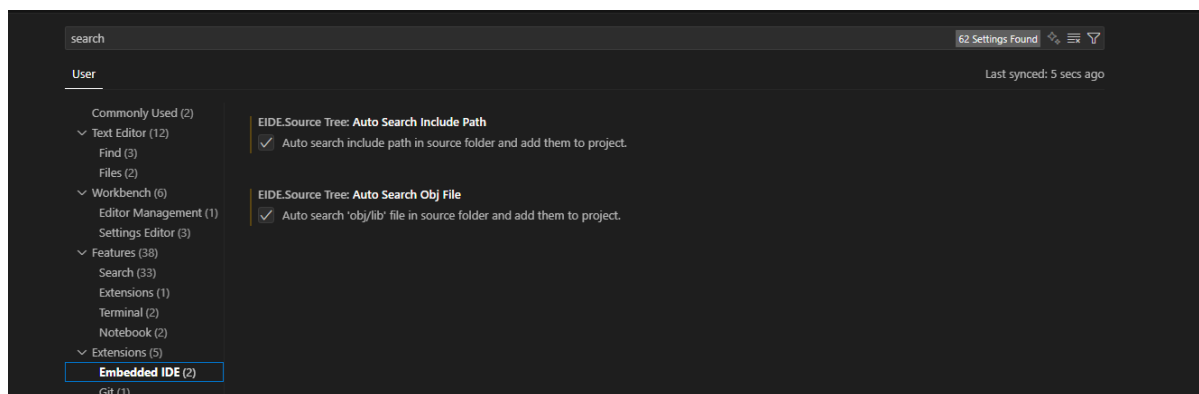
最后还有一个要勾选的：点击左下角的设置



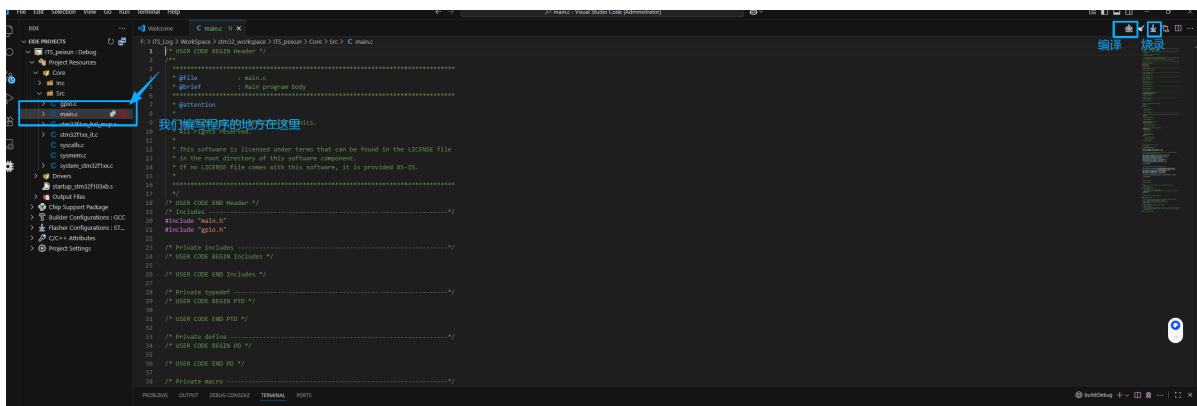
在框内输入search, 然后点击Embedded IDE



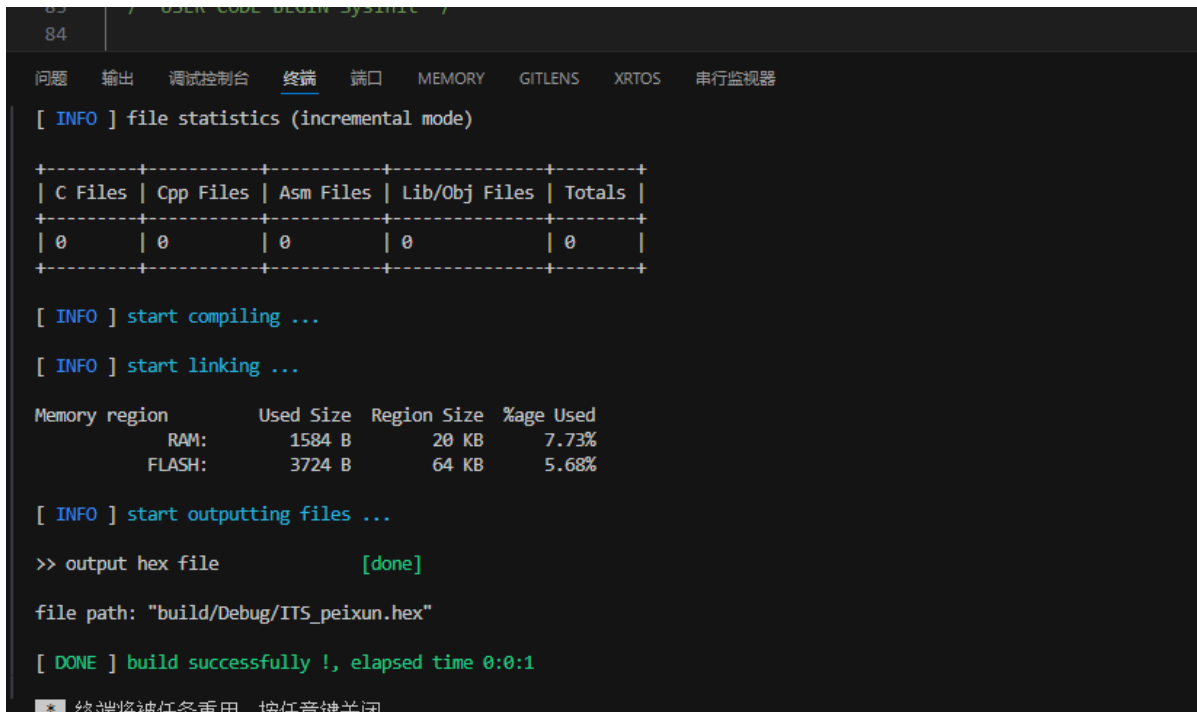
保证这两个都勾上了（其实第一个就是刚才添加的），本次主要是添加第二个



配置至此告一段落啦



编译一下验证是否配置正确:



在战队拿完板子可以进行烧录:

